Capstone Project 1

By

Abishek S H

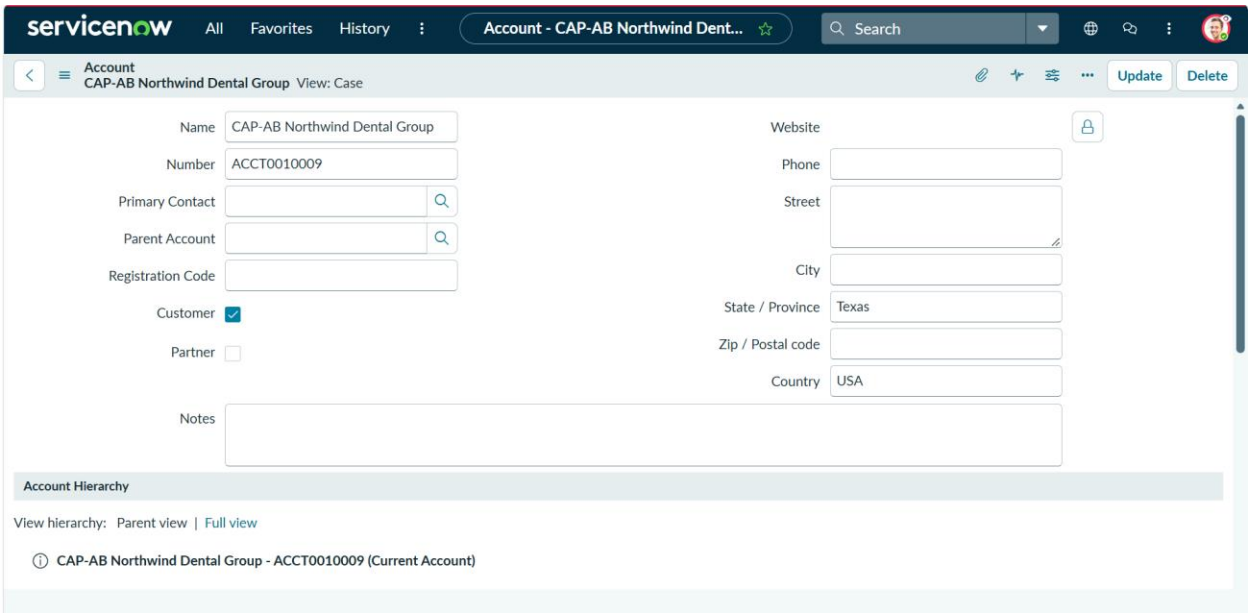
PART A: Customer and case

A1: Customer setup

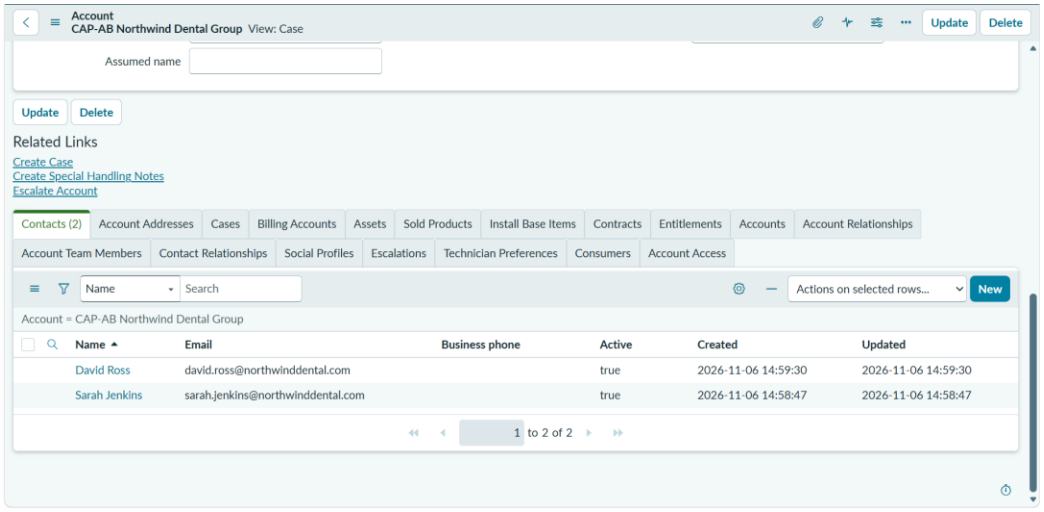
Requirement: Create the customer account CAP-XX Northwind Dental Group with two contacts that have different roles (a practice manager and an IT lead). Create a service contract for the account, with an entitlement that allows the customer to raise device support cases.

Evidence:

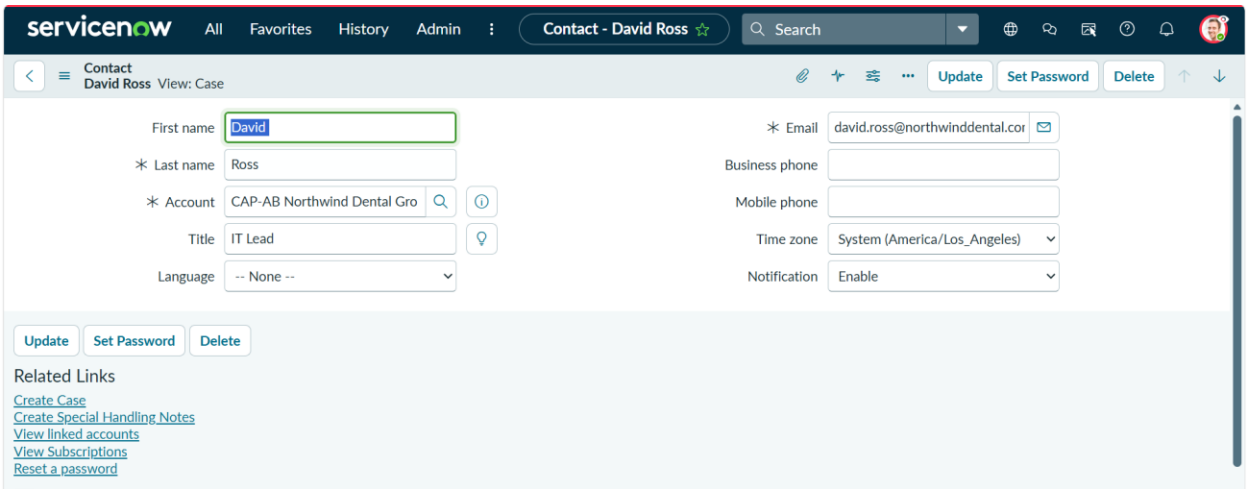
Account created for CAP-AB Northwind Dental Group



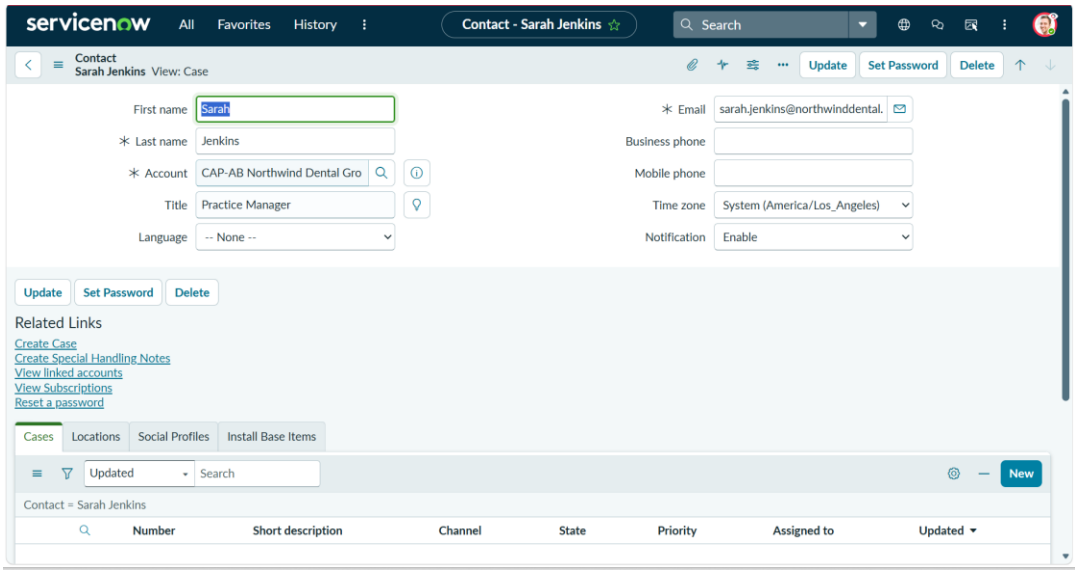
Two contacts created for the account:



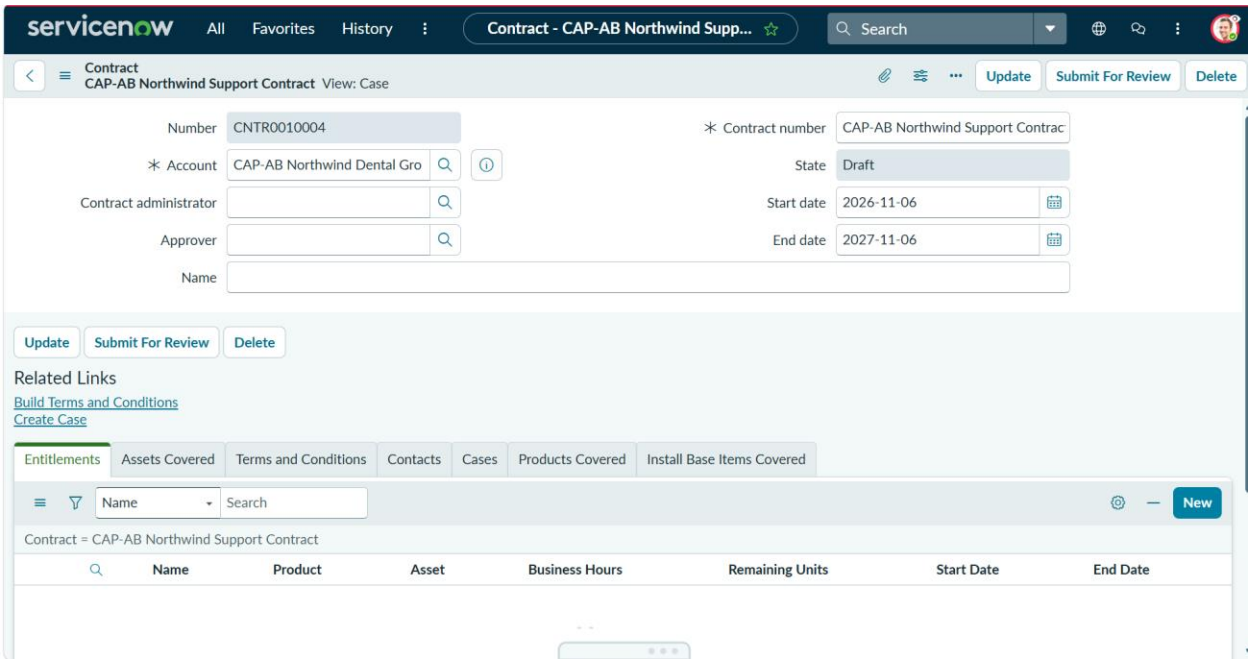
First contact: David Ross



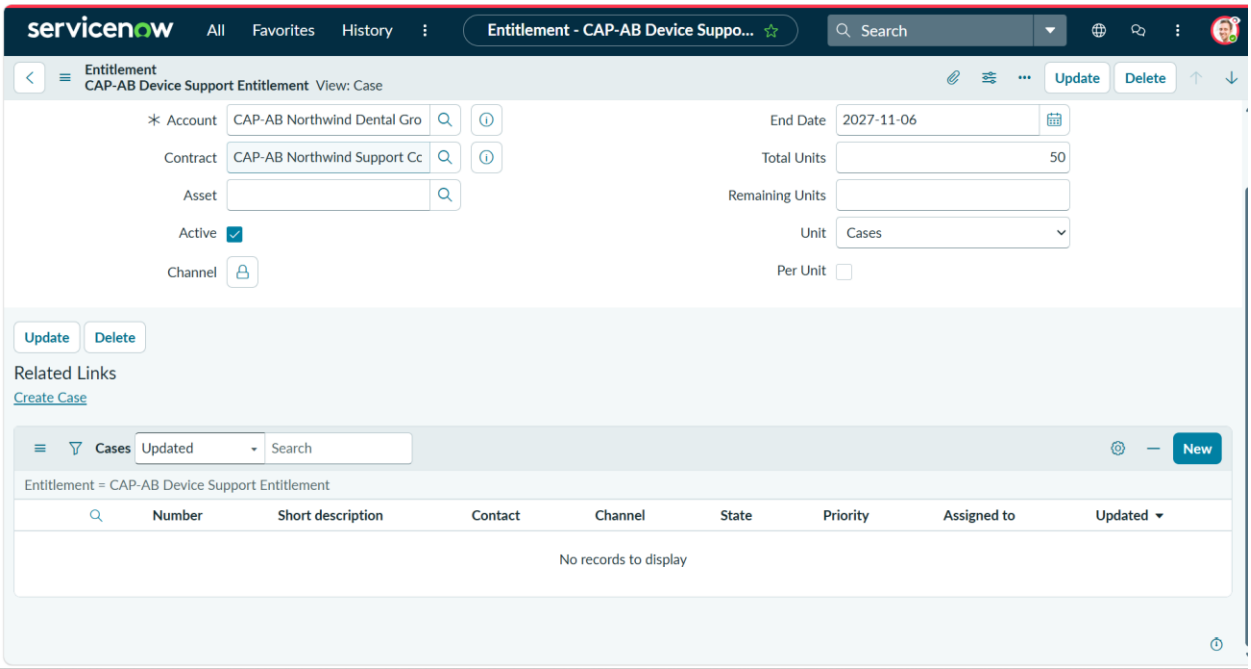
Second contact: Sarah Jenkins:



Contract created for the account:



Entitlement created for the account:

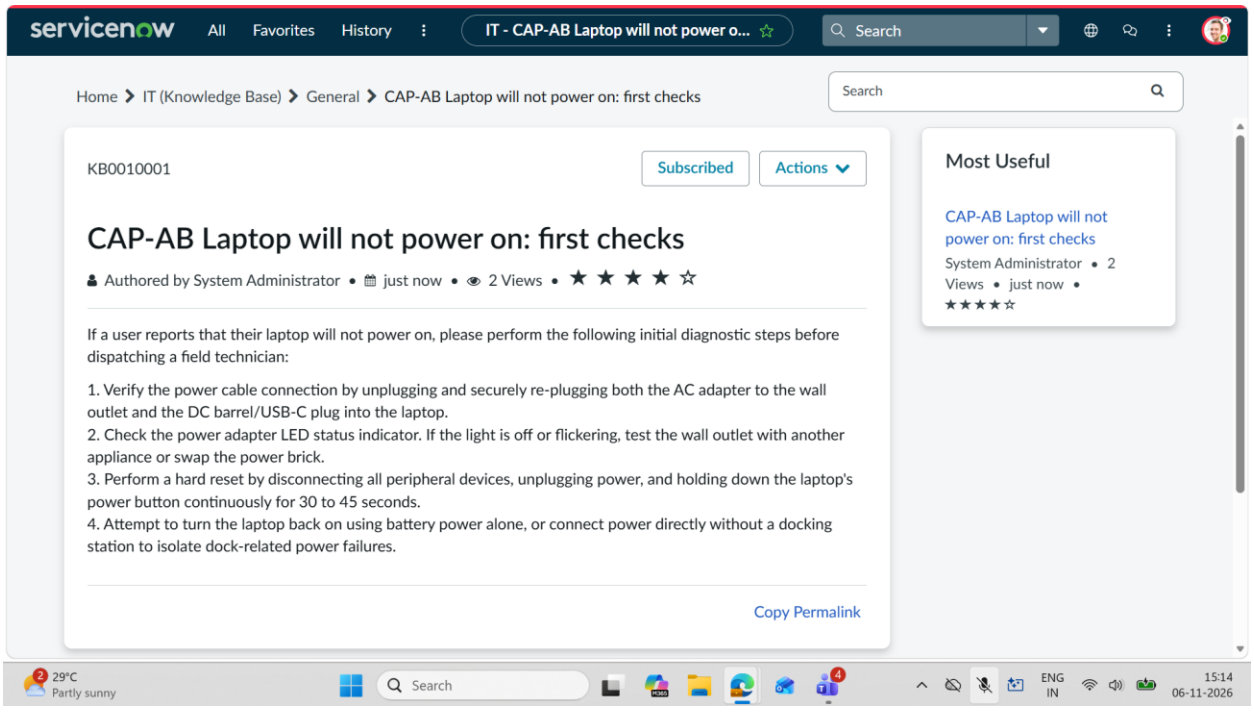


A2: Knowledge Article

Requirement: Create and publish a knowledge article titled CAP-XX Laptop will not power on: first checks. It must contain at least four numbered troubleshooting steps and be usable by customer service agents.

Evidence:

Knowledge base published in the instance:



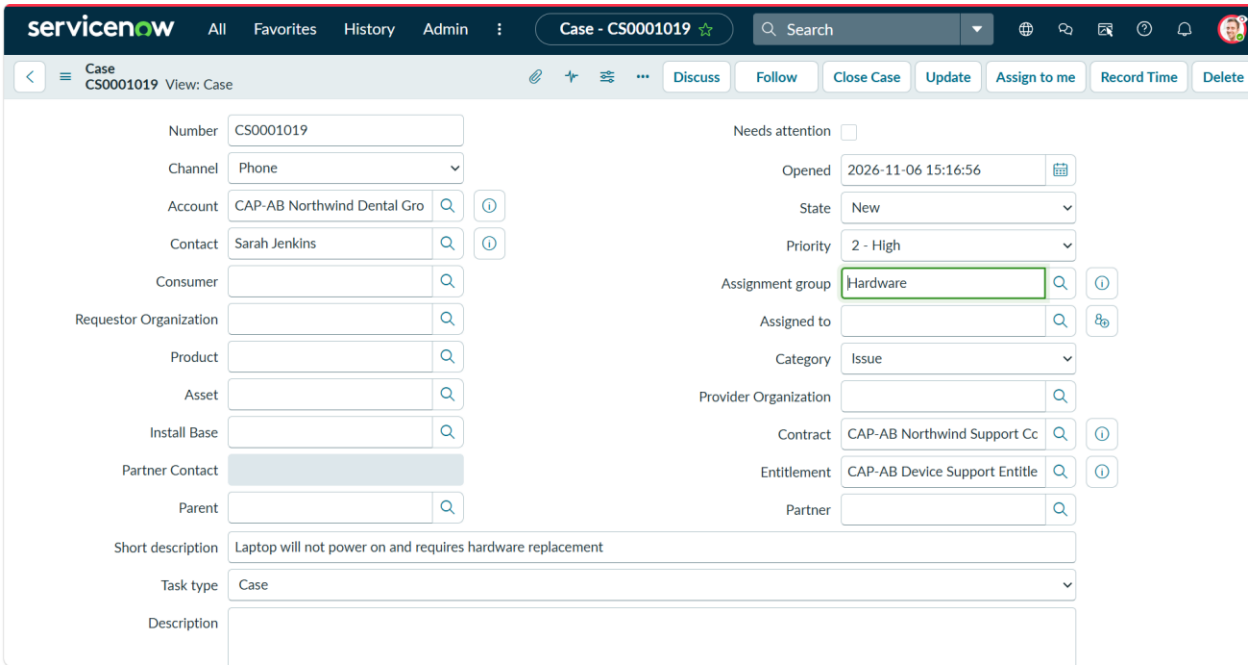
A3: Case from first contact to resolution

Requirement: Create a case for the practice manager that reports the laptop failure. Categorize and prioritize it appropriately, assign it to a suitable group, attach the knowledge article, and make sure the case is covered by the entitlement and has an SLA. Progress the case through its lifecycle to Resolved after the field visit in Part B is complete.

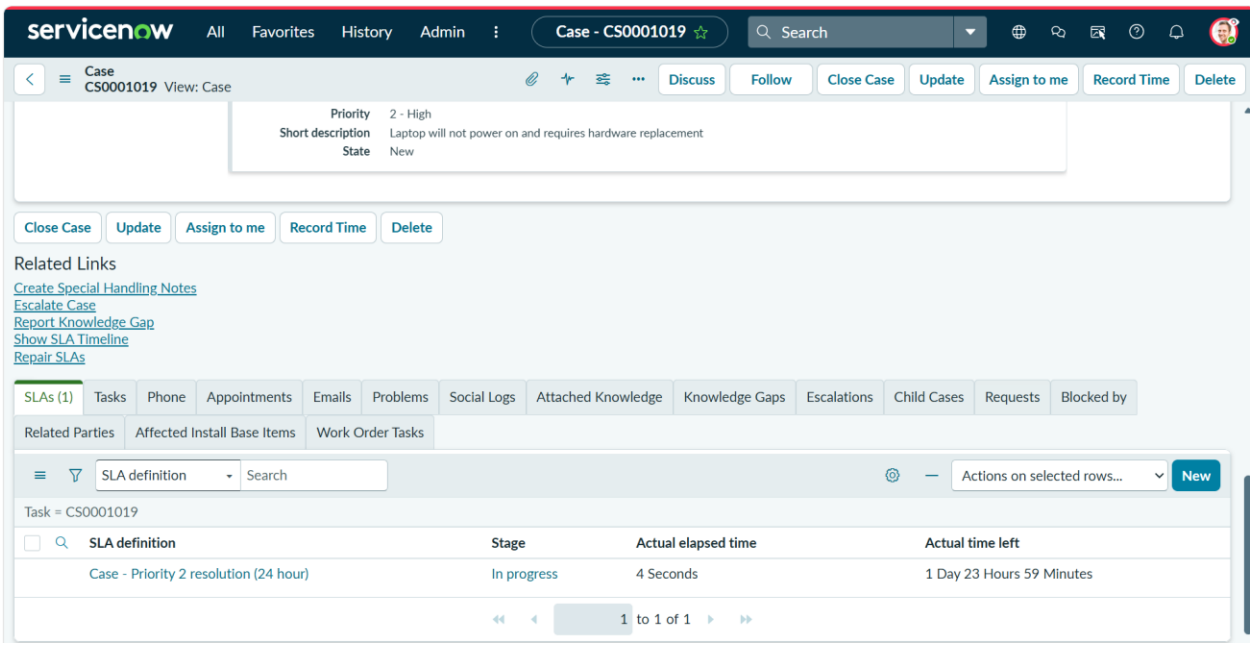
Evidence:

Case created with the contact associated with the account.

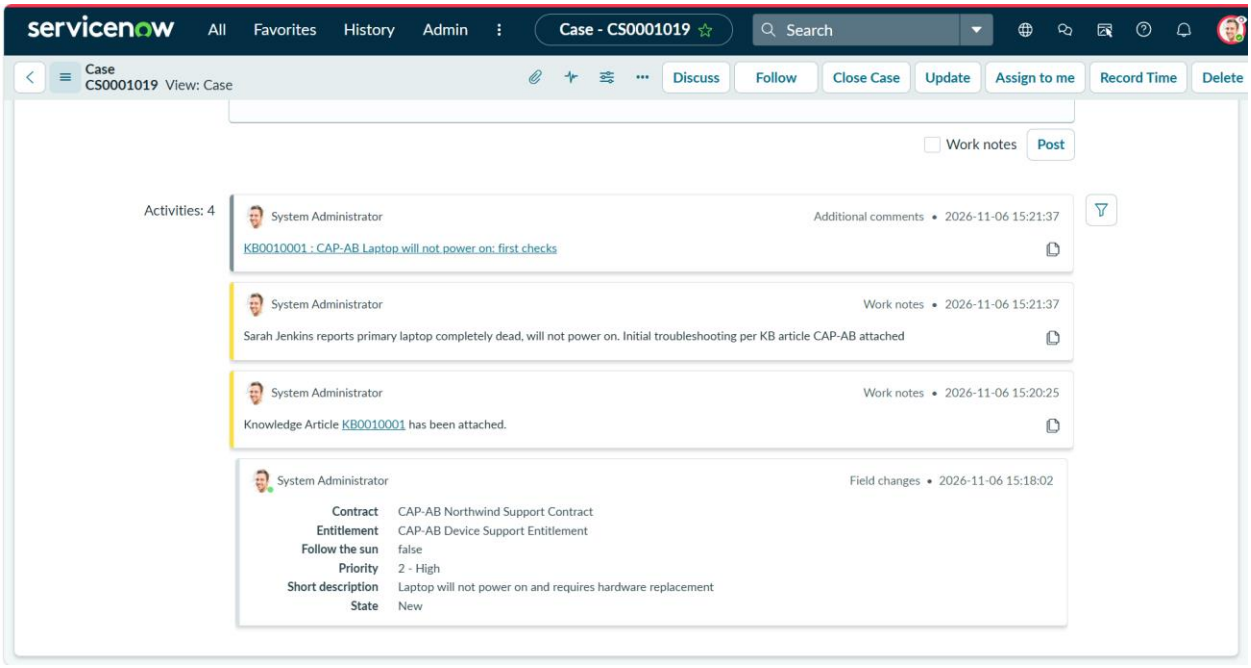
Contract and entitlement are auto populated based on the account.



SLA has been attached for the case:



Knowledge article attached along with the work notes for the case:

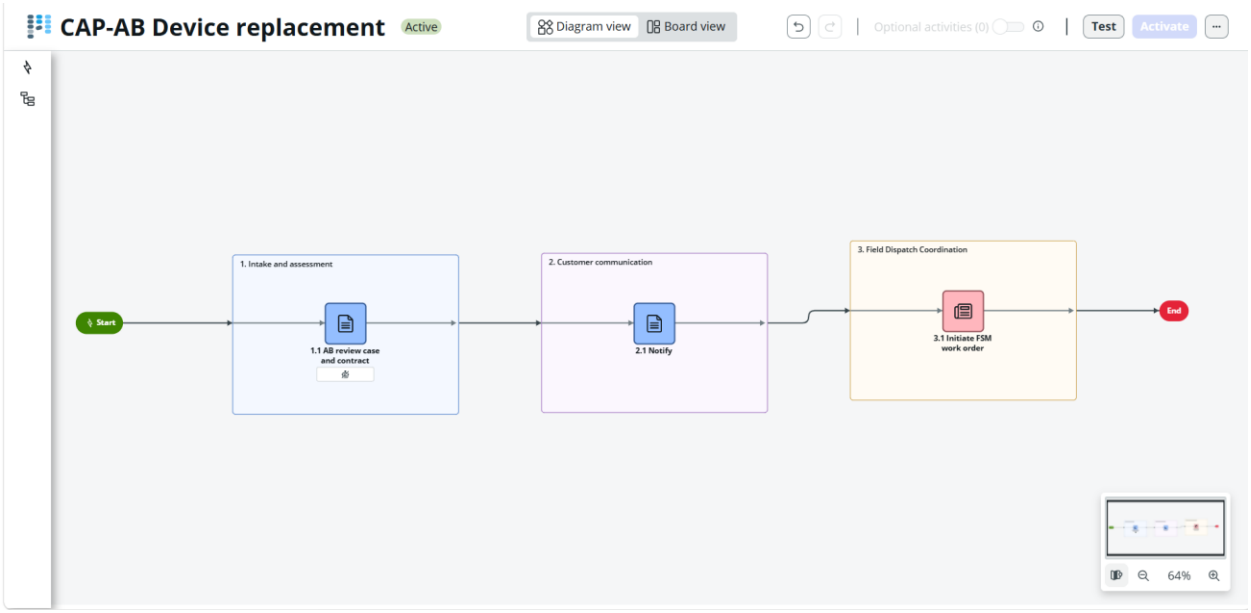


A4: Simple Playbook

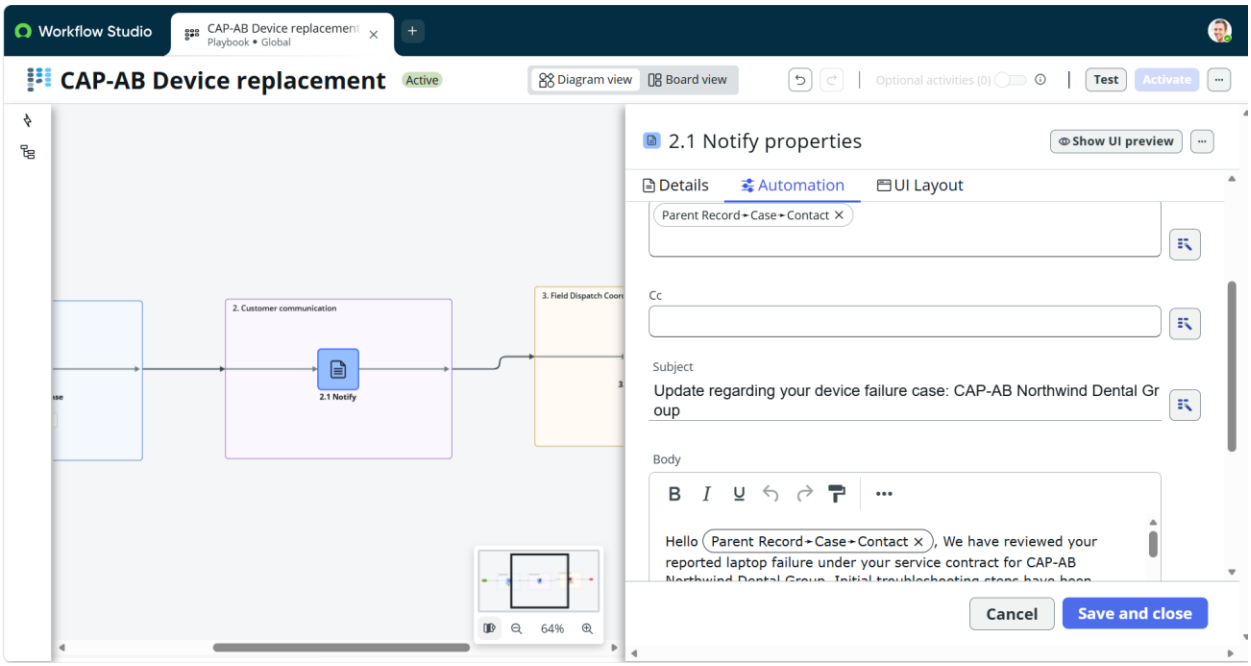
Requirement: Configure a playbook named CAP-XX Device replacement for cases about device failure. It must have at least three steps, including one manual step for an agent and one step that communicates with the customer. Show that the playbook is used on, or can be triggered for, your case.

Evidence:

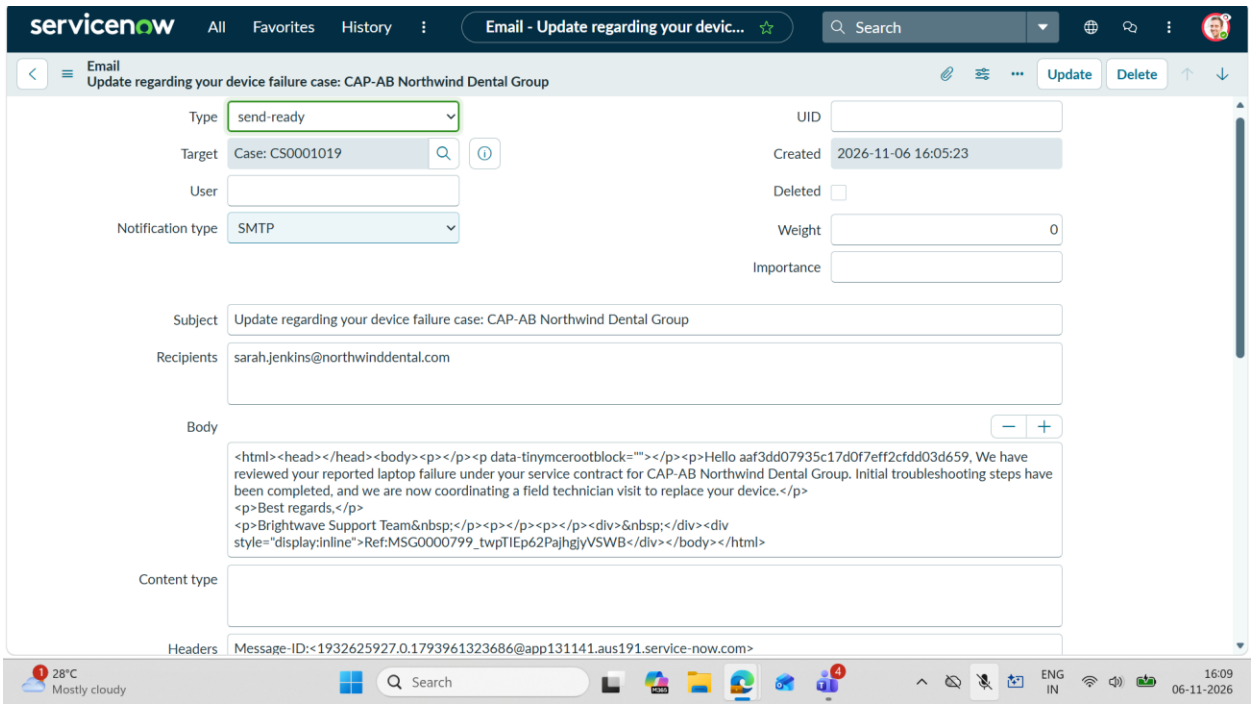
Playbook created:



Automation stage for communication:



Automation email sent to the contact:



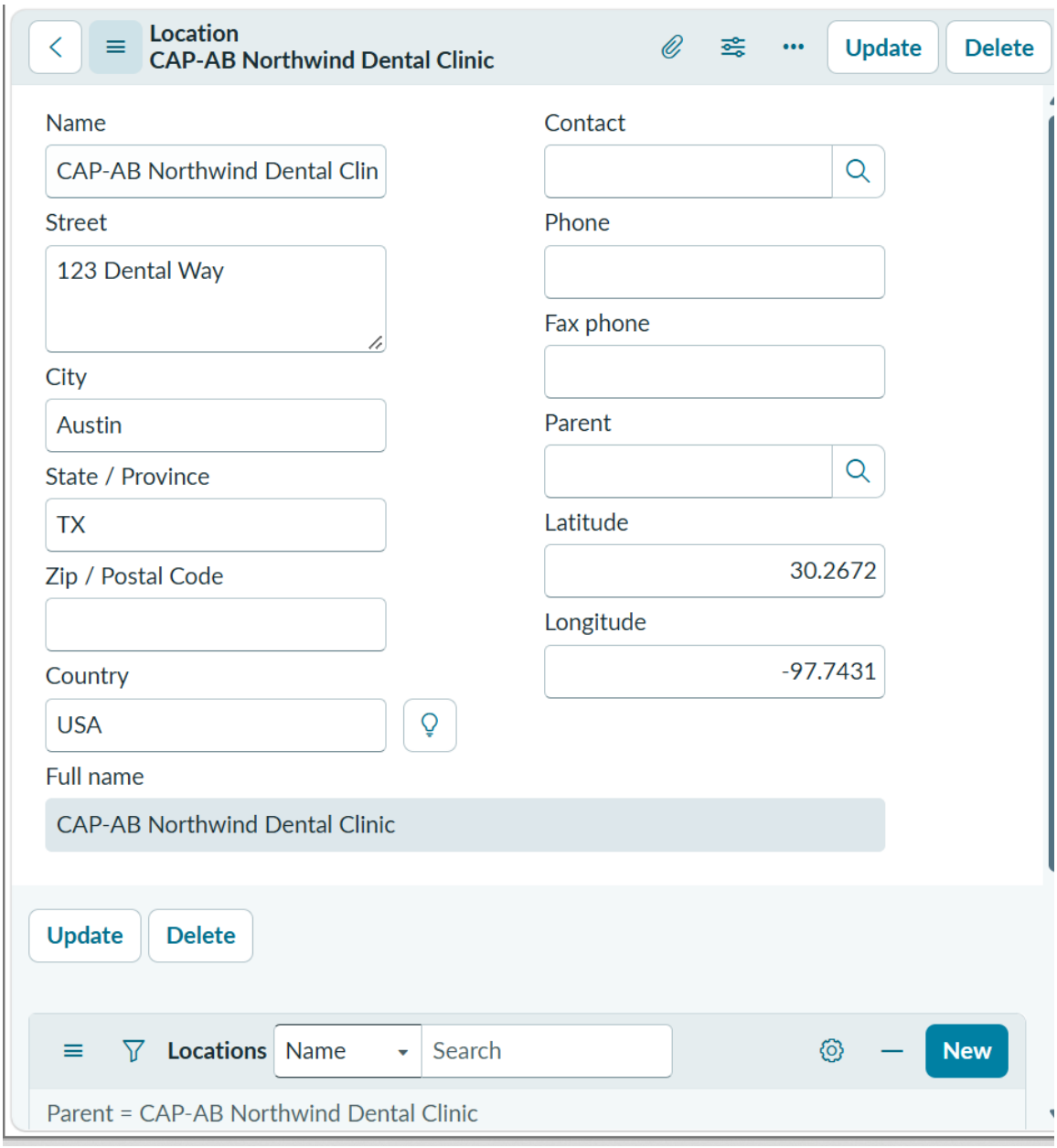
Part B: Field Visit

B1: Work Order from the case

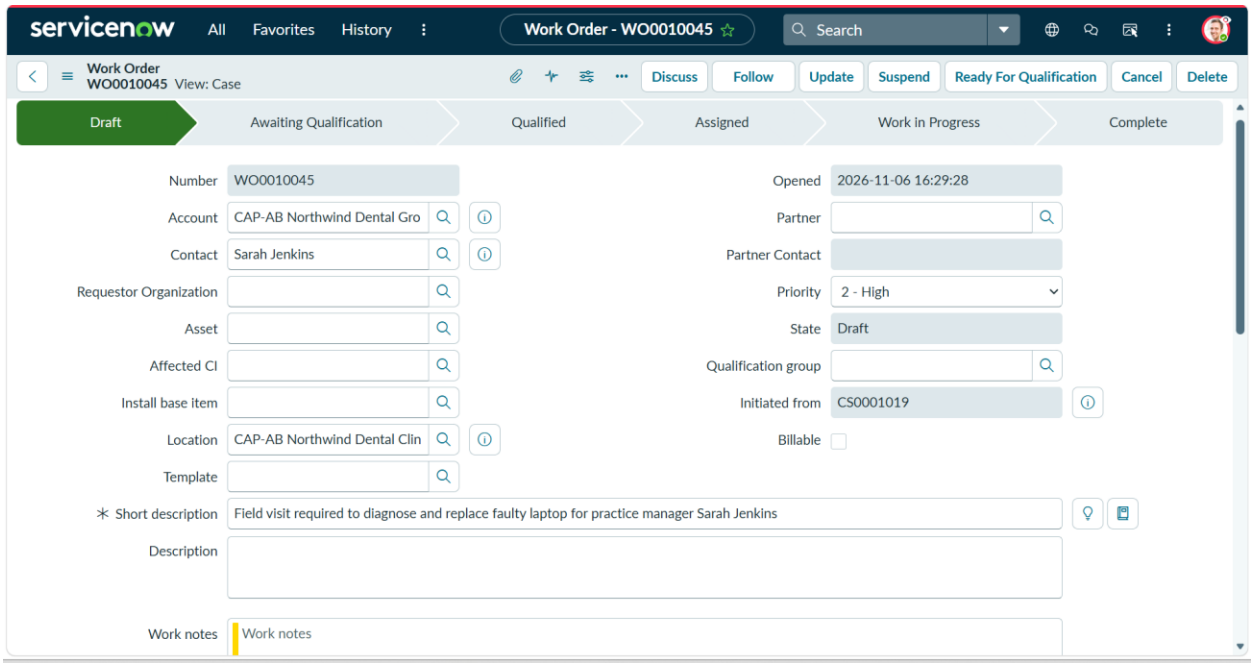
Requirement: Create a work order for the field visit from your case. It must show the customer, a service location (create a location named CAP-XX Northwind Dental Clinic), a clear description, and the link to your case.

Evidence:

Location created:



Work order created from the case:

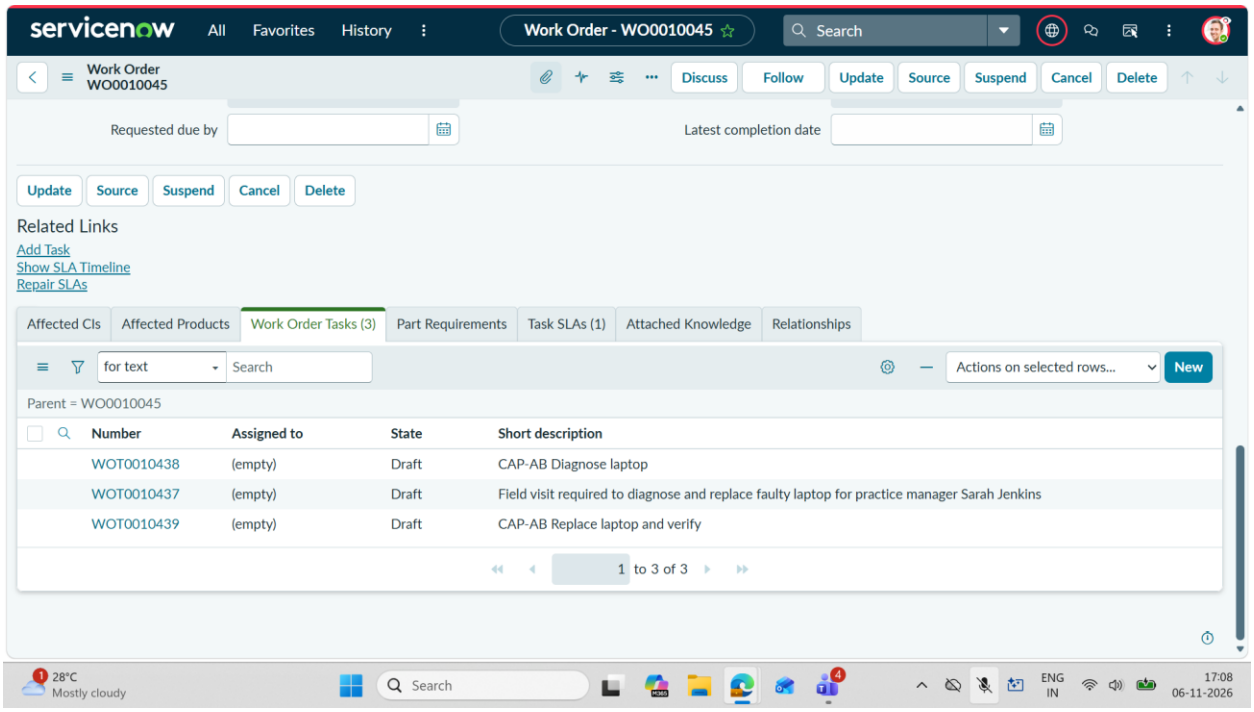


B2: Tasks, skills and dispatch

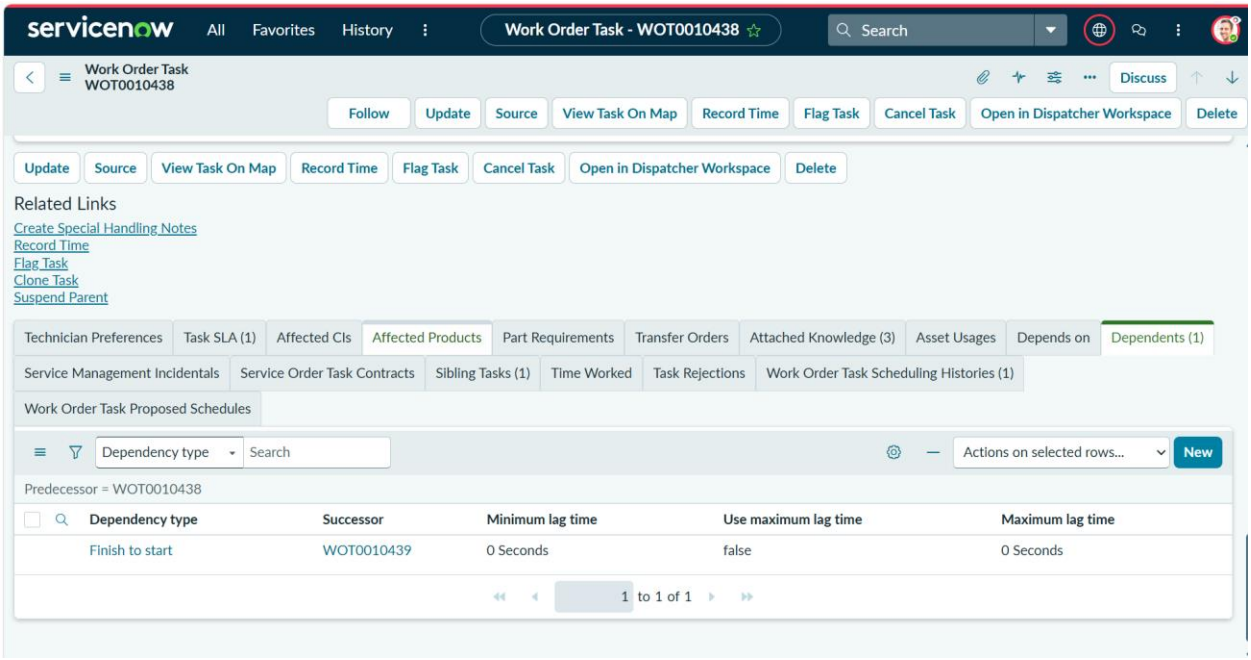
Requirement: Add two tasks to the work order: CAP-XX Diagnose laptop and CAP-XX Replace laptop and verify. The second task must not start before the first one is finished. Each task needs at least one required skill (existing or created, named with CAP-XX). Create one technician with the matching skill and assign both tasks to that technician. Take both tasks and the work order to completion.

Evidence:

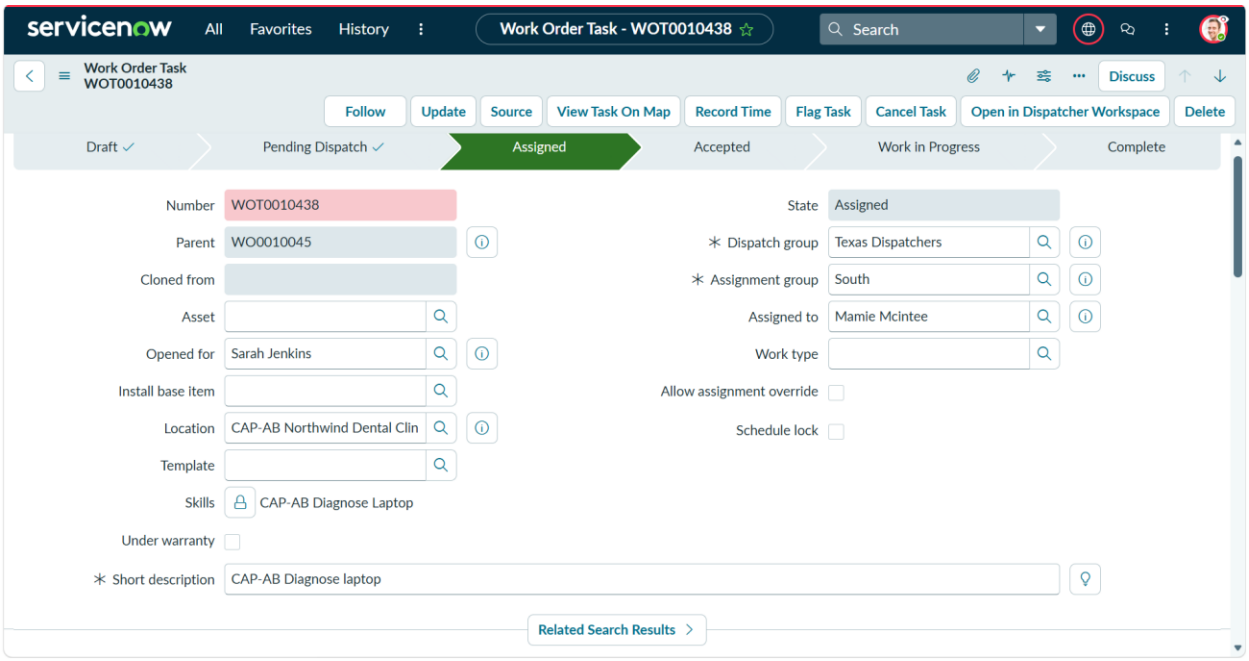
Work order tasks created:



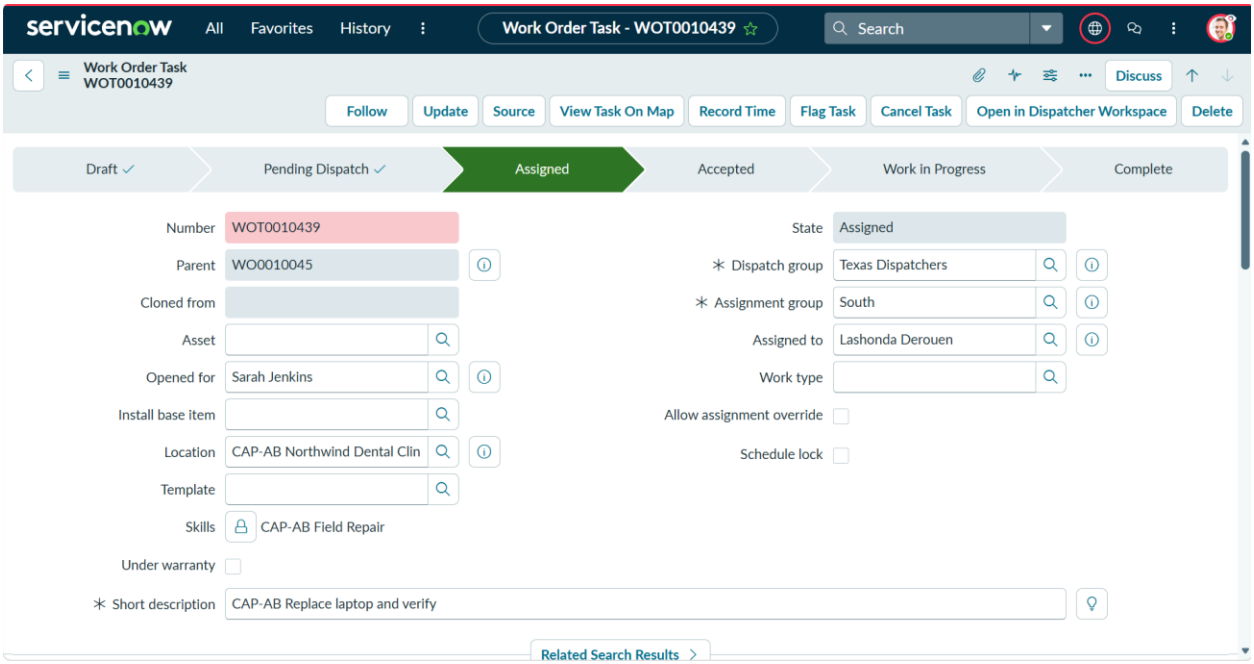
Dependency of work order task:



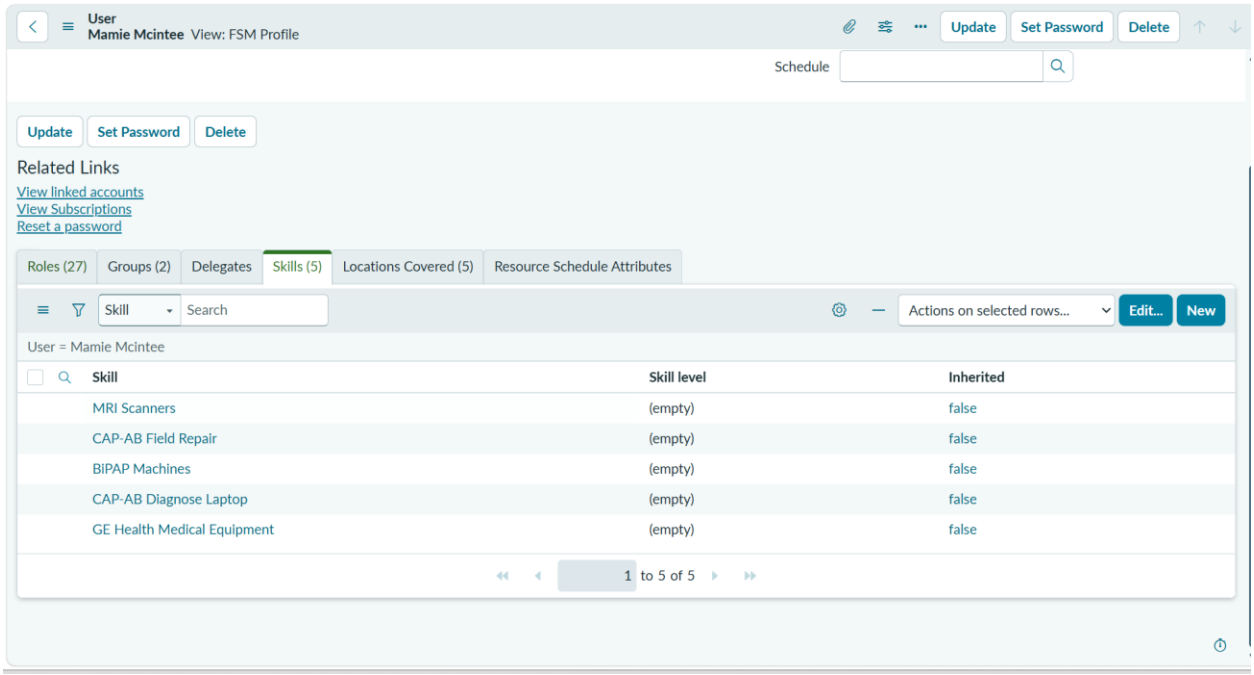
Work Order Task 1 Assigned:

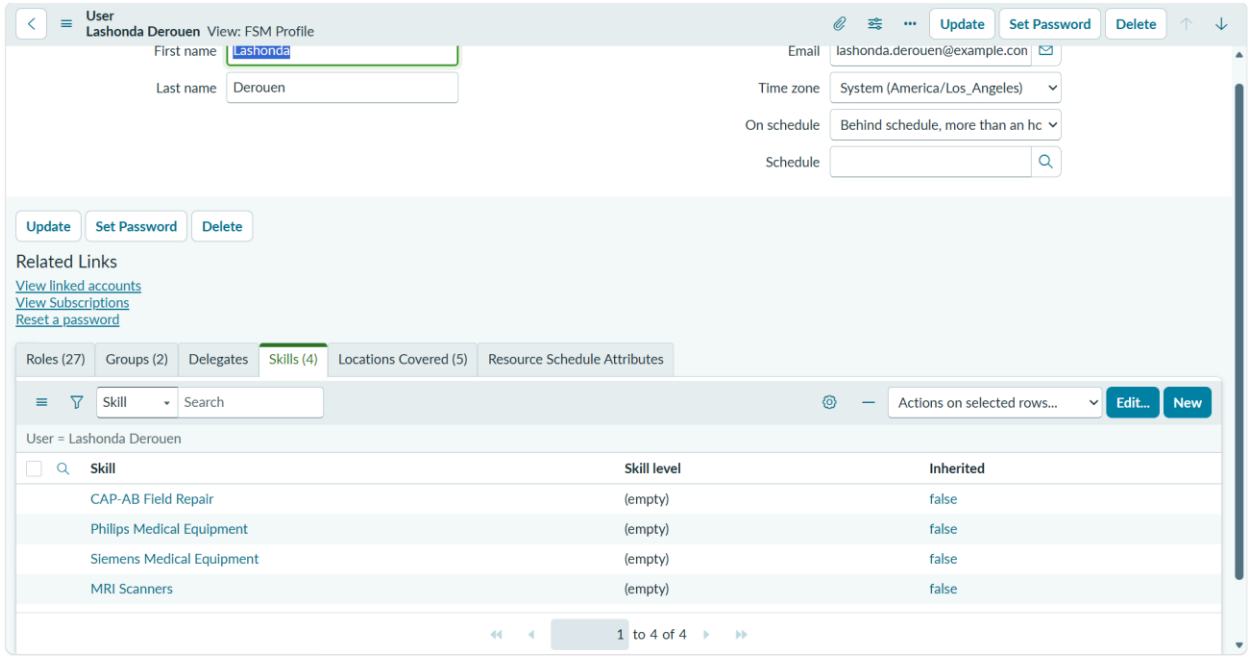


Work Order Task 2 Assigned:

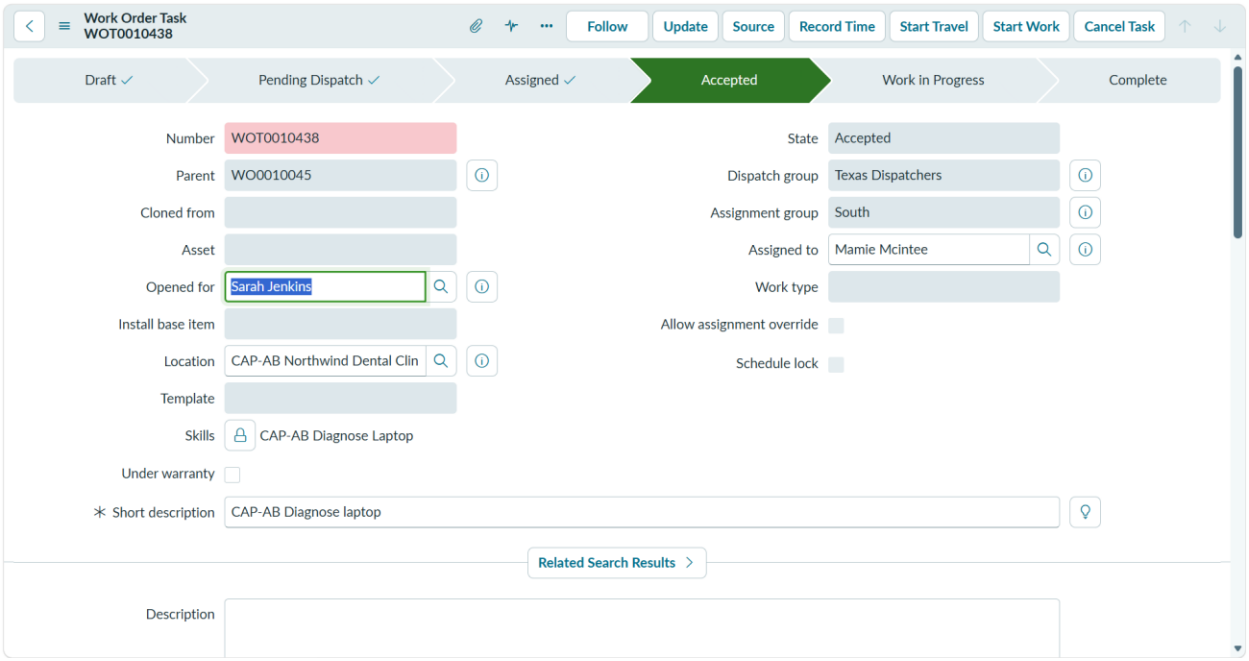


Skills for the Work Order has been added to the Technicians:

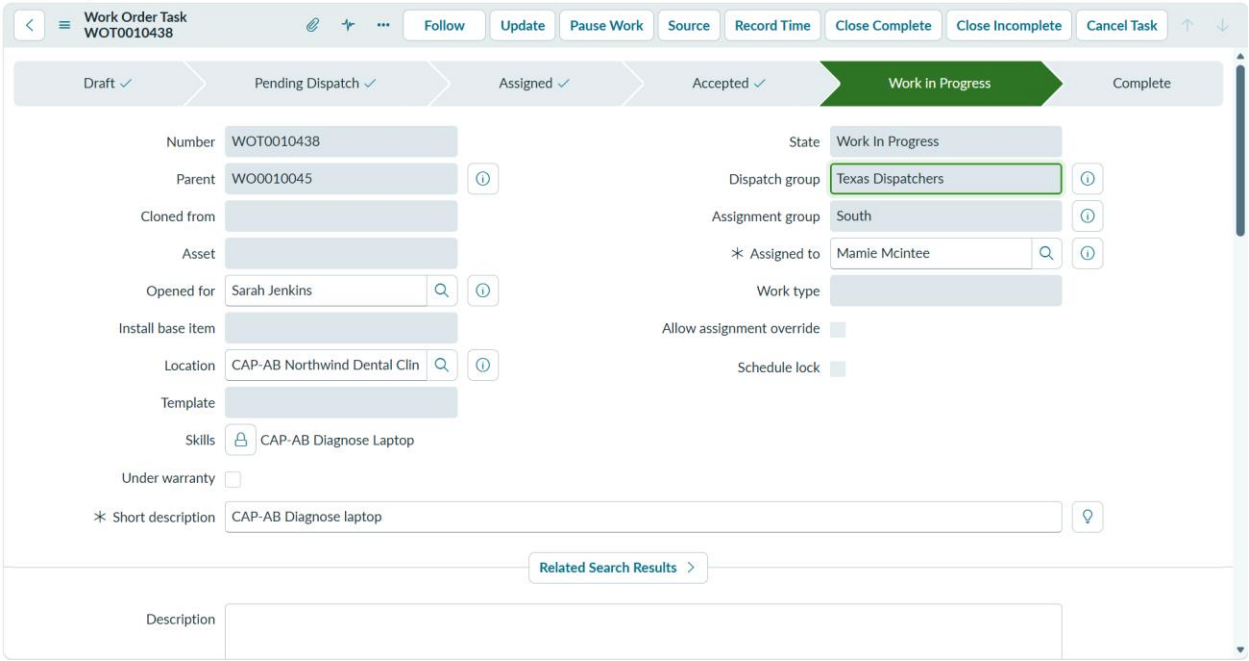




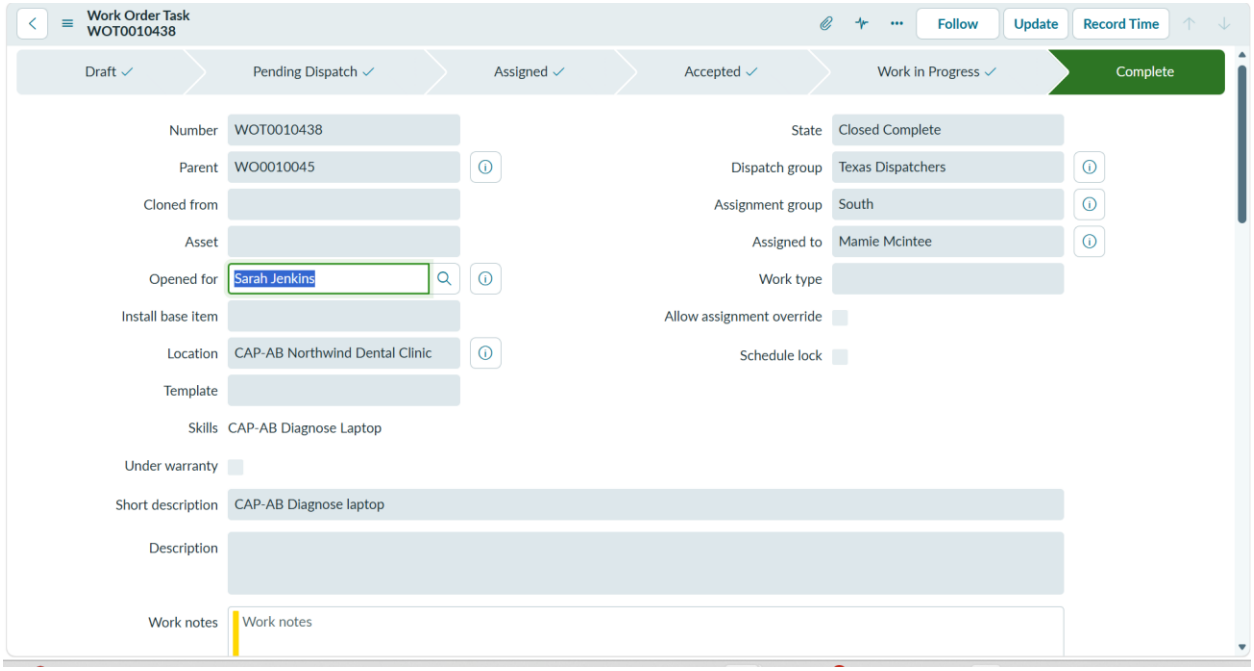
Technician accepted the work order task:



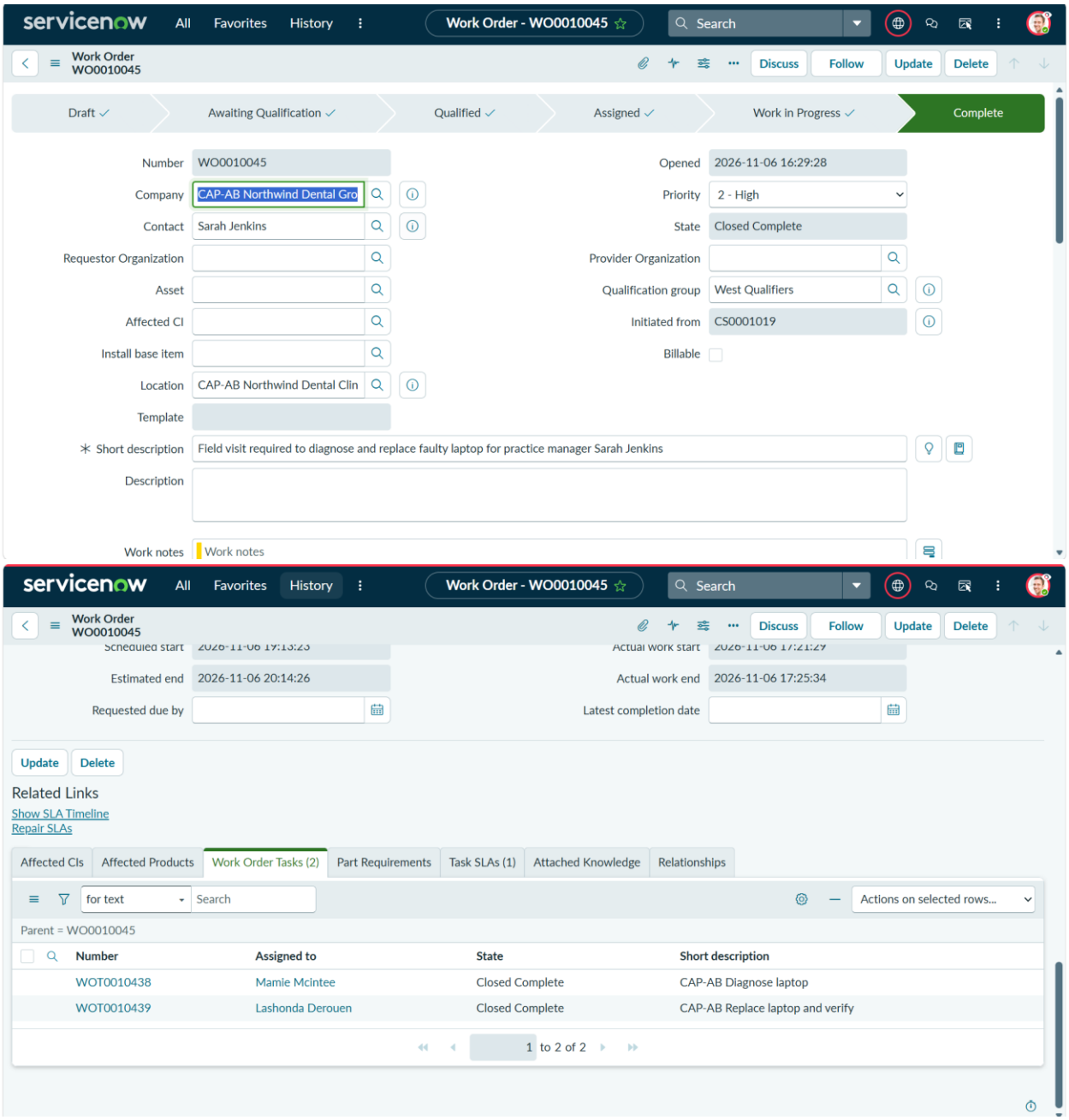
After start travel and start work done by Technician:



After providing the work notes and close complete the work order task:



Work Order state completed after all the work order task is closed completed:

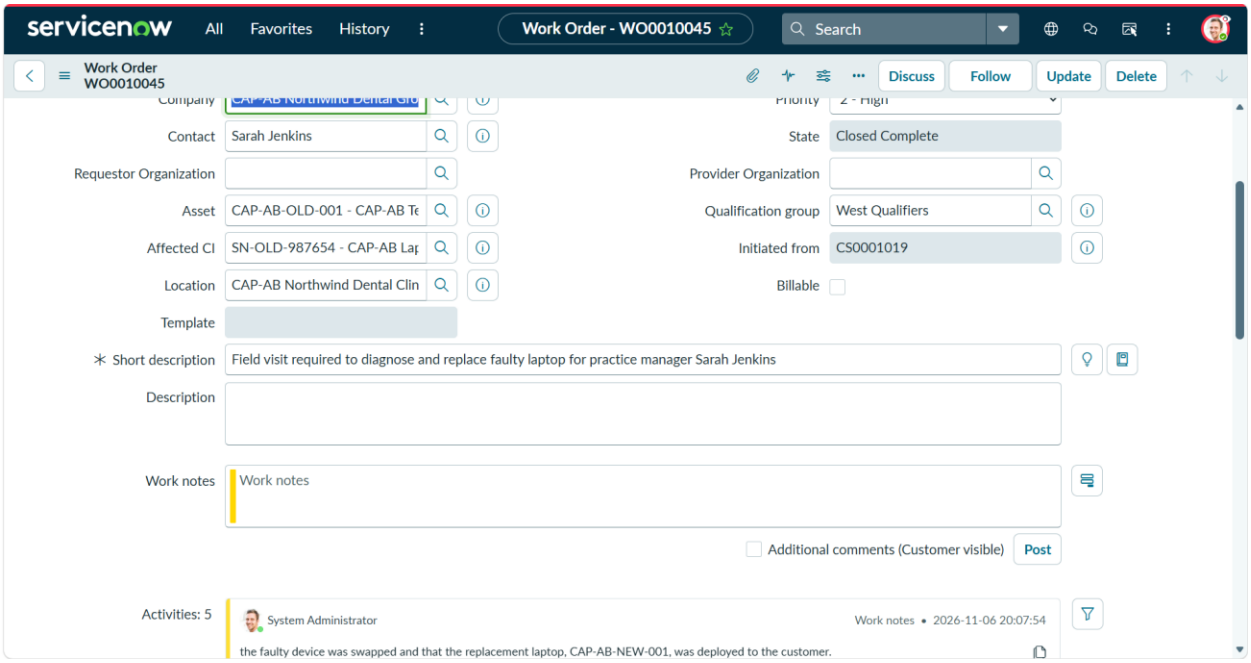


B3: Device on the work order

Requirement: After completing Part C, show on the work order the device it concerned: record the faulty laptop as the affected asset

or CI, and the replacement laptop in the work notes or in the field your configuration provides for it.

Evidence:



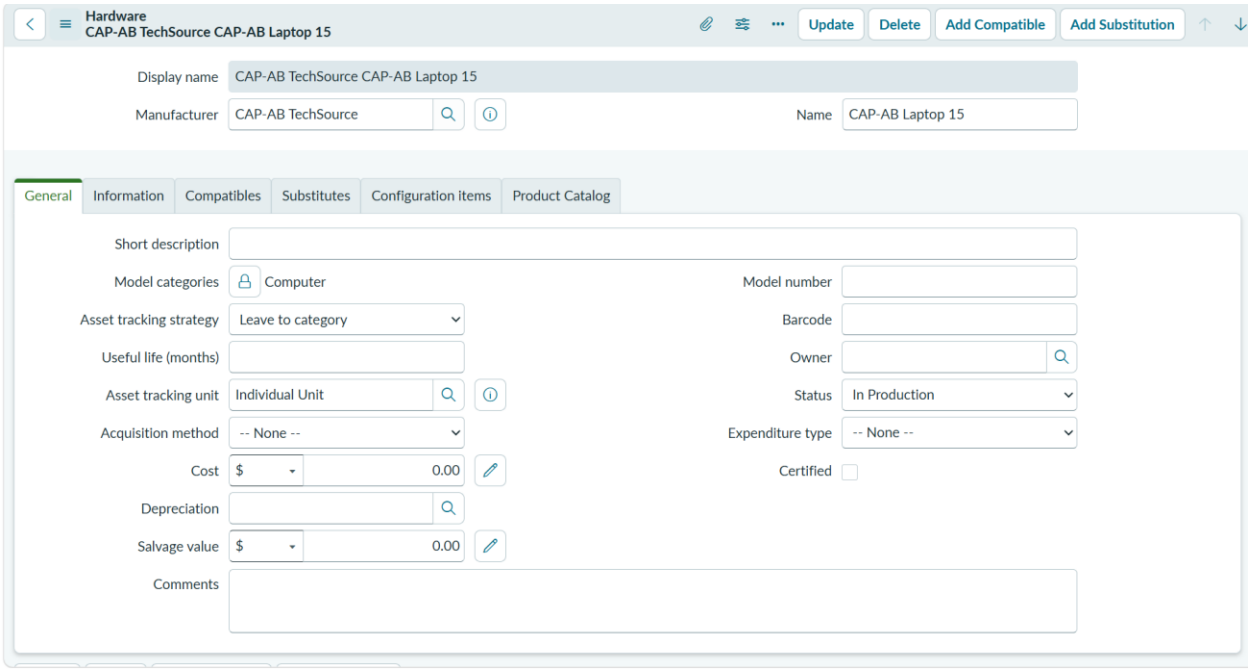
Part C: Assets and CMDB

C1: Master Data

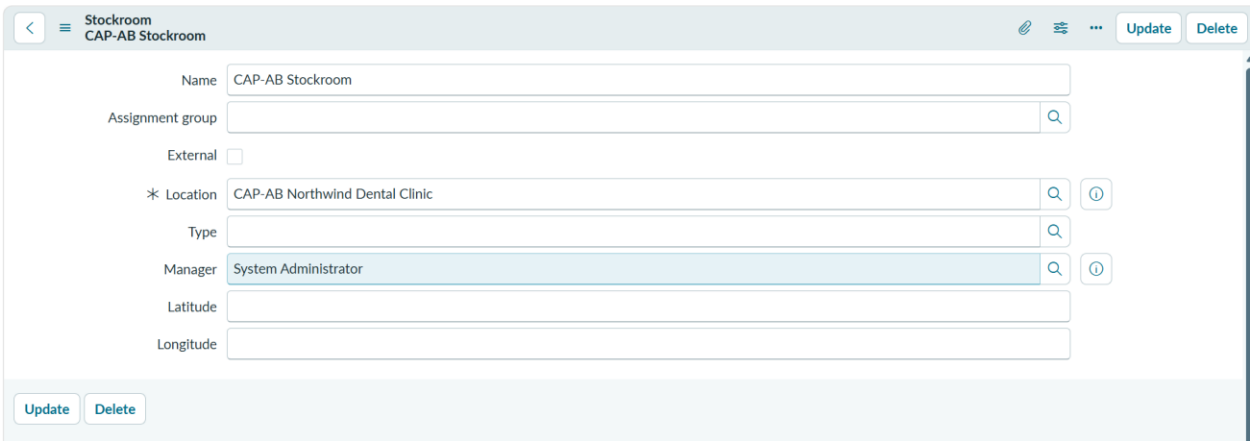
Requirement: Create a manufacturer CAP-XX TechSource, a hardware model CAP-XX Laptop 15 in the Computer category, and a stockroom CAP-XX Stockroom with a manager.

Evidence:

Product Model:



Stockroom:

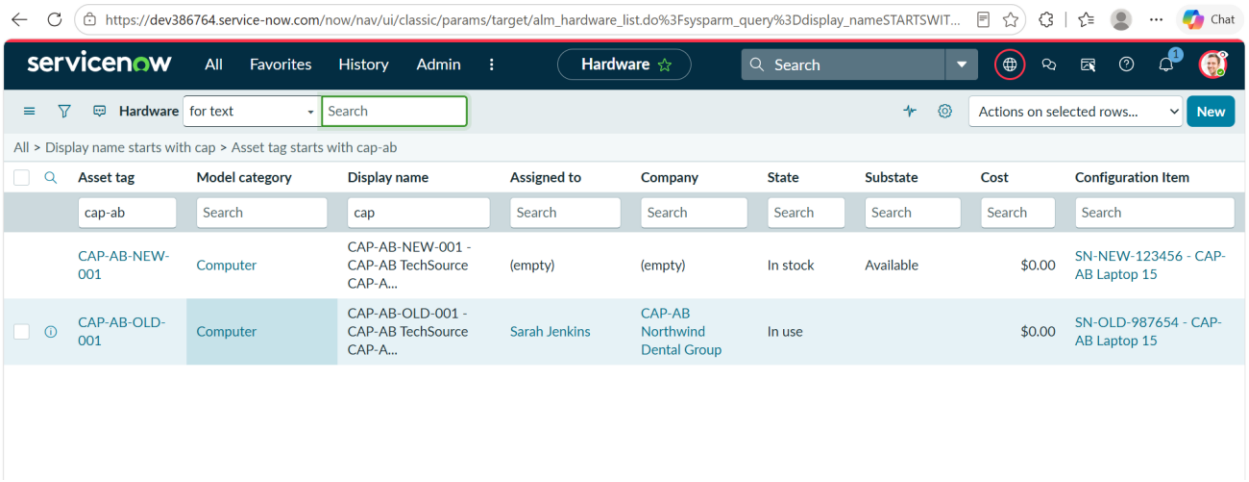


C2: Two Laptops

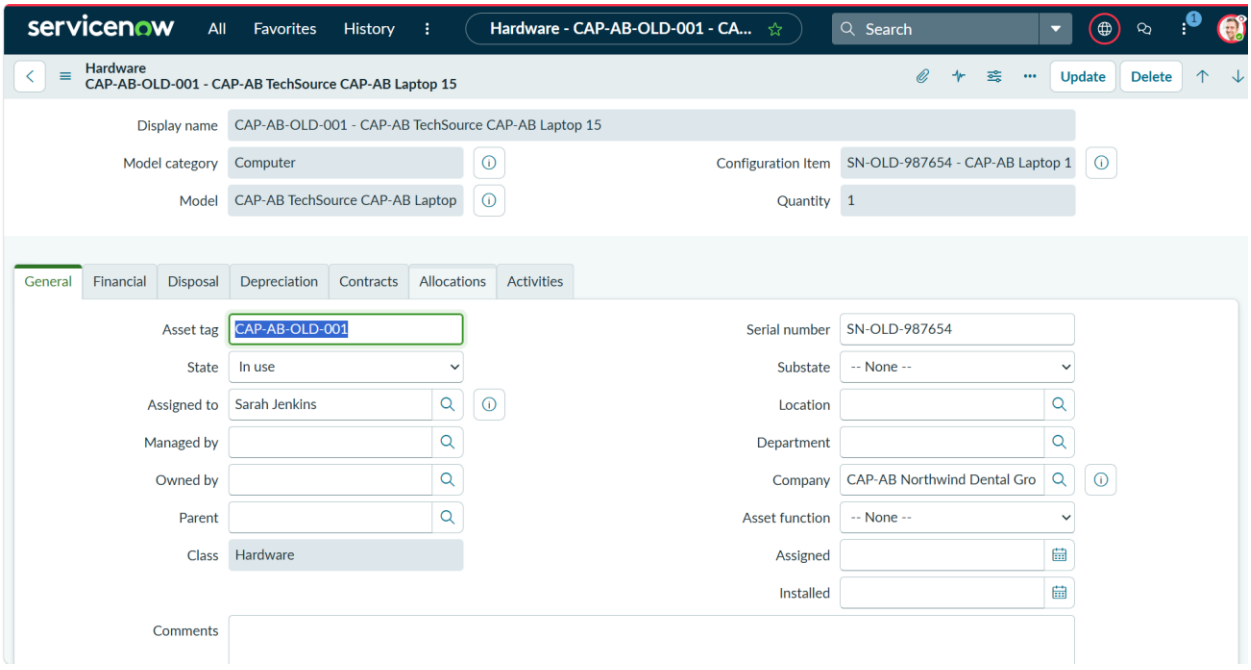
Requirement: Create two hardware assets from your model: the faulty laptop (tag CAP-XX-OLD-001) and the replacement laptop (tag CAP-XX-NEW-001). Give each a unique fictional serial number. At the start of your scenario the faulty laptop is in use by the customer contact and the replacement is waiting in your stockroom.

Evidence:

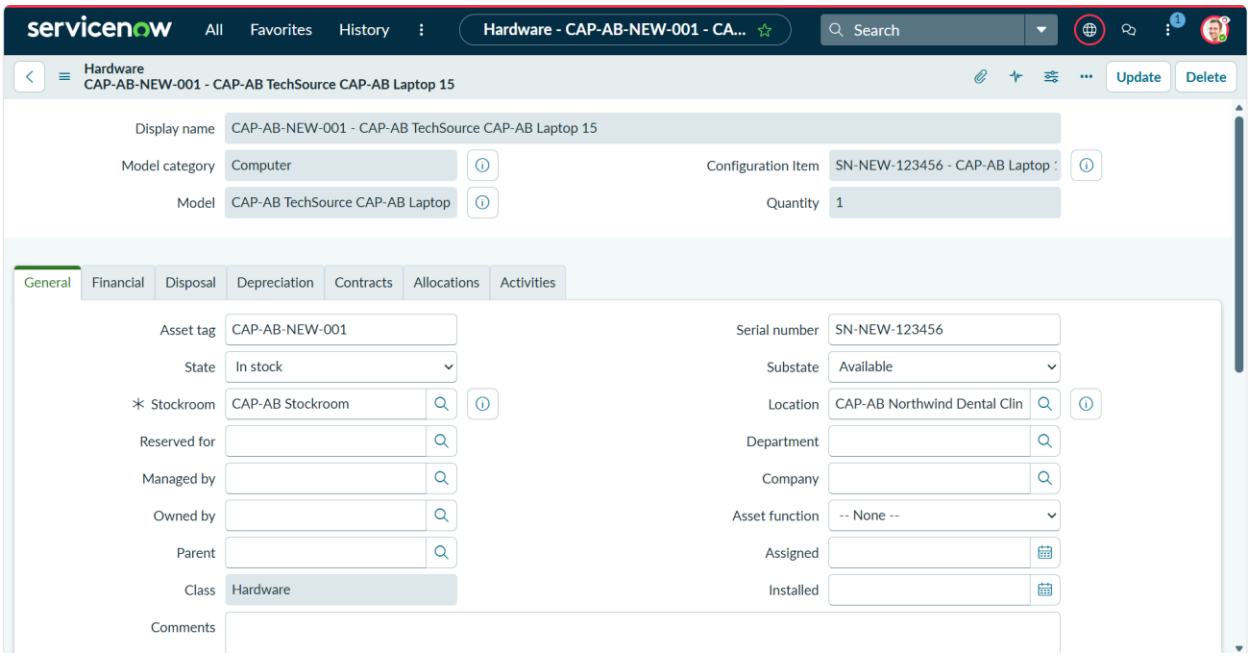
Filter in the hardware for Laptops created:



Faulty Laptop:



New Laptop:

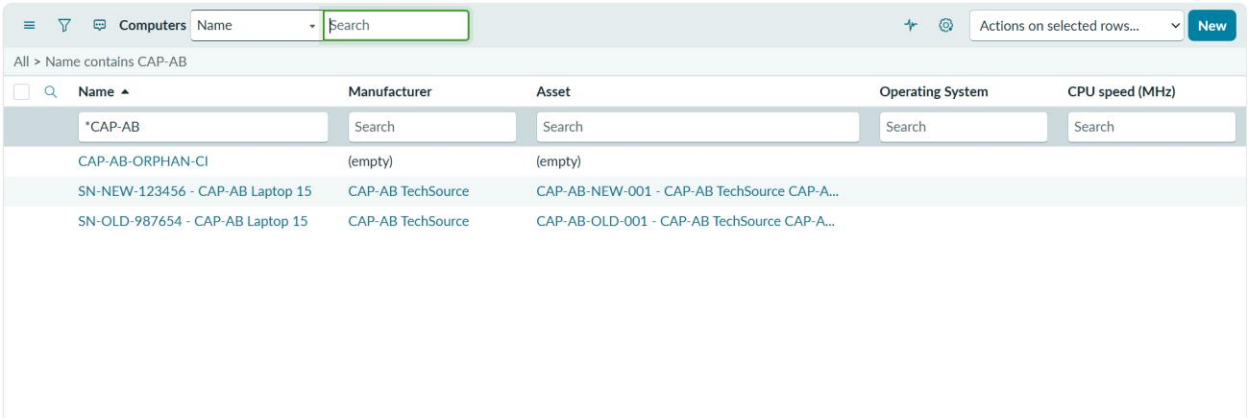


C3: CI Link and Reconciliation

Requirement: Make sure each laptop has a matching CI linked to it. Then create one deliberate mismatch of your choice (an asset without a CI, a CI without an asset, or two CIs with the same serial number). Detect it with a list filter, then resolve it. Use the serial number as the match key.

Evidence:

Orphan CI created without asset:

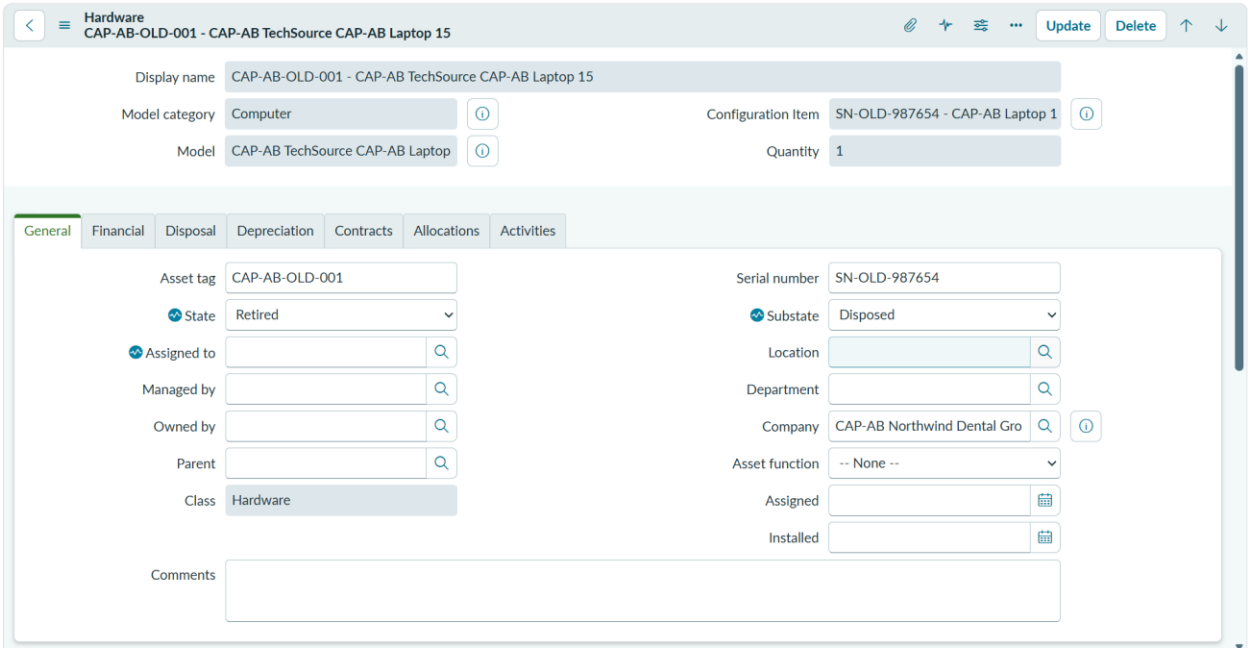


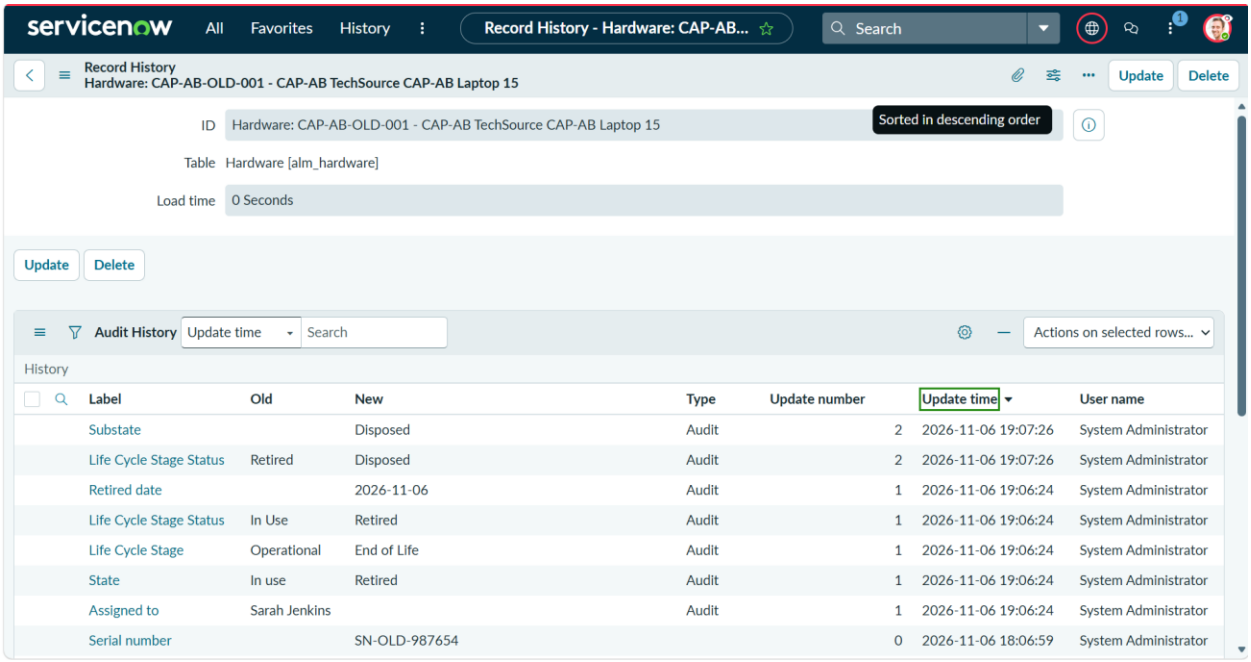
C4: Replacement lifecycle

Requirement: Complete the device swap in the asset records: deploy the replacement laptop to the contact and return and retire the faulty laptop. The asset states and fields must tell a correct story.

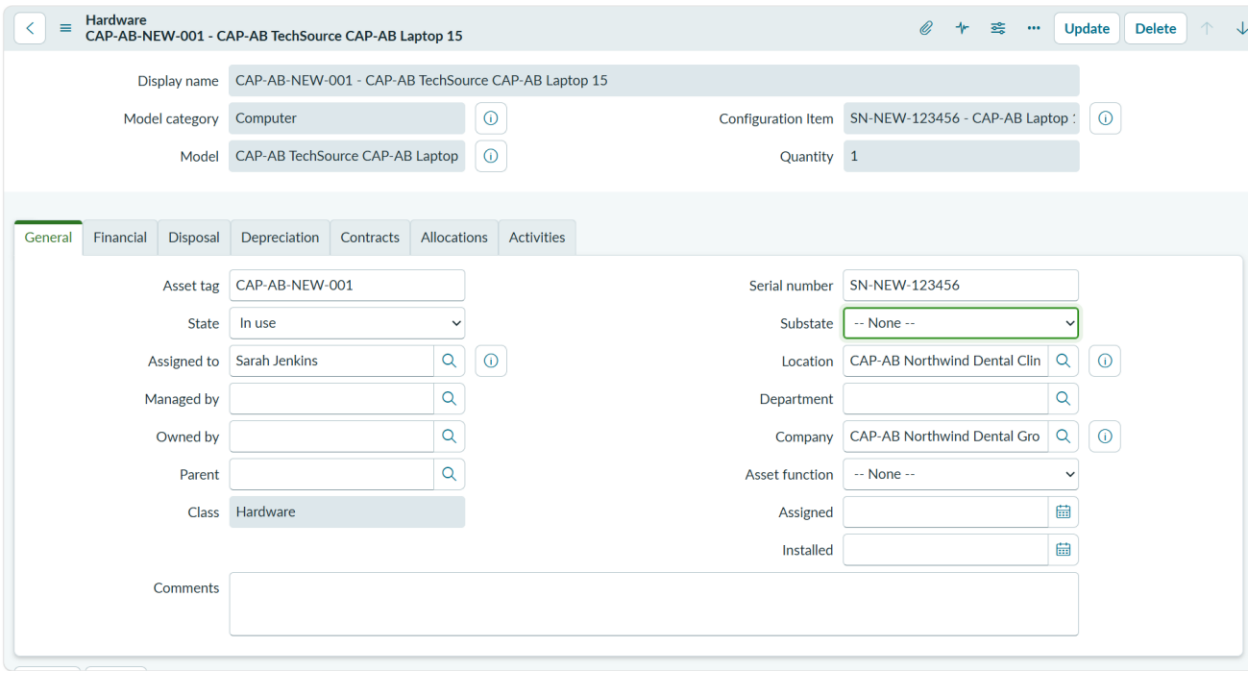
Evidence:

Old Laptop state changed to Disposed:





New Laptop assigned to Sarah Jenkins:



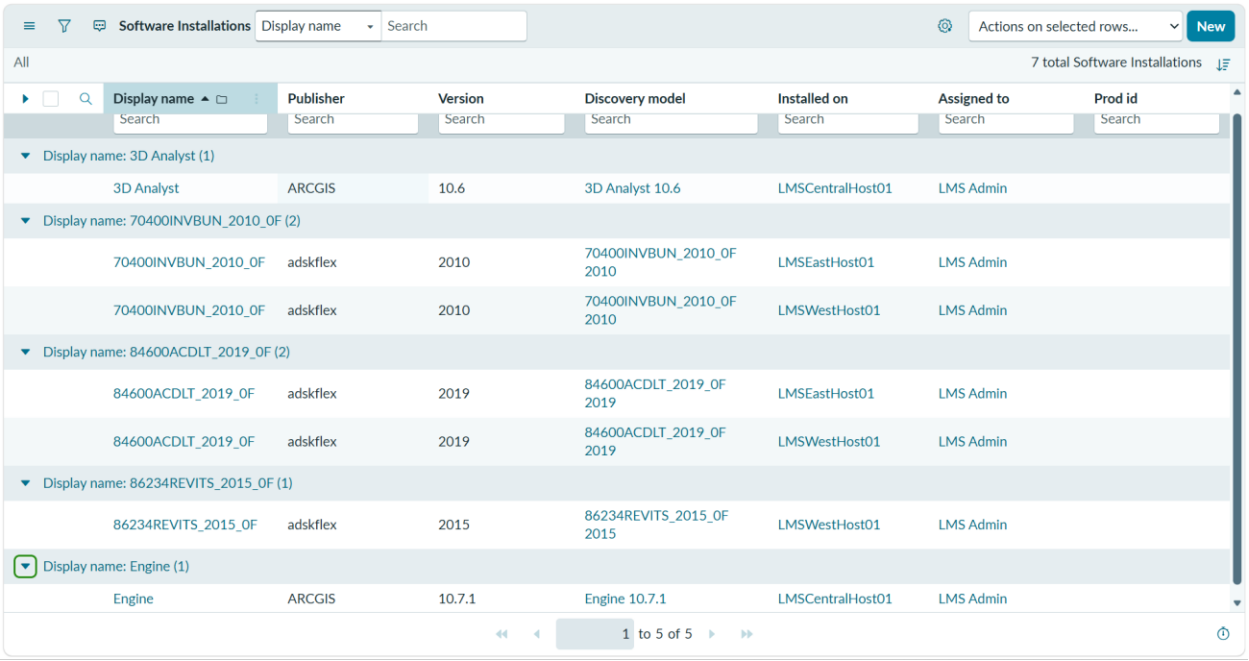
Part D: Software position

D1: Installation

Requirement: Pick one software product that has installations in your instance's software installation data. Record the product, its publisher, the number of installations, and the names of at least three computers that have it. Choose a product with at least five installations if one exists.

Evidence:

Filter of software installation:

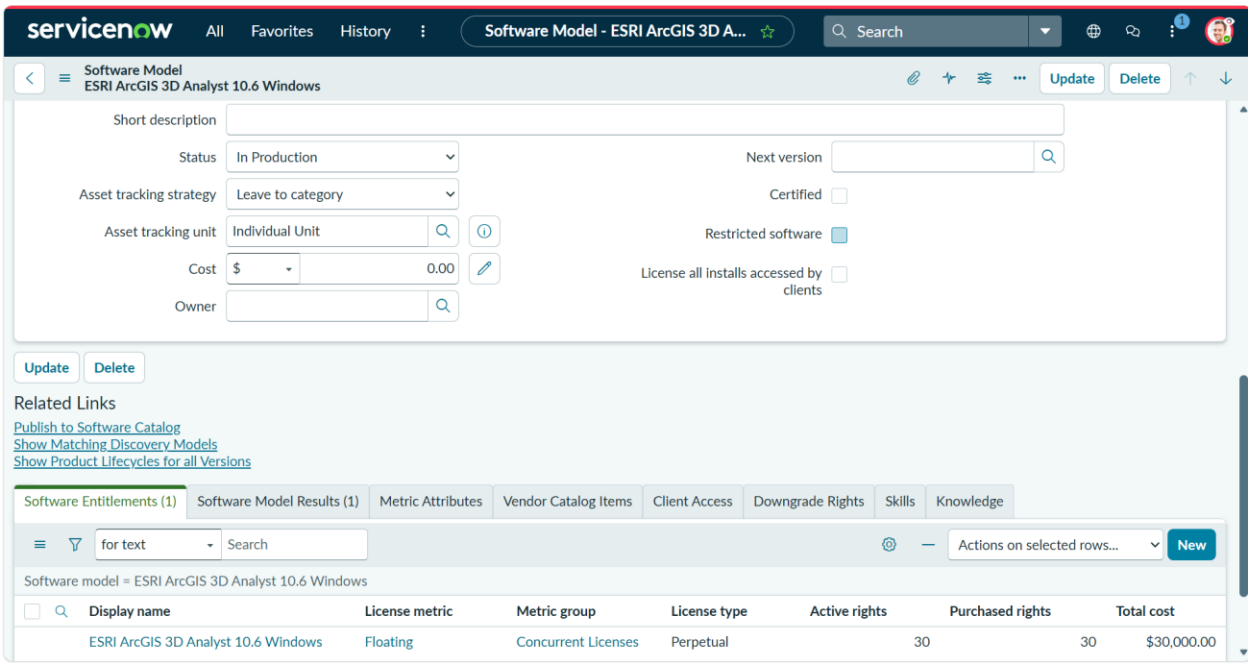


D2: Entitlements and recommendations:

Requirement: Record rights for that product by creating an entitlement against a software model for the product, using the licence metric Per Device and purchased rights that are three fewer than your installation count (or one if the count is three or fewer). Use a price of 100 per right as an assumption. Calculate the licence position and write a recommendation.

Evidence:

Software Entitlements of the software model:



Insights:

Software Model: ESRI ArcGIS 3D Analyst 10.6

License Metric: Per Device (or Concurrent / Floating as shown in your entitlement row)

Total Installations: 30 installations

Purchased Rights: 30

License Position: Compliant (installation and purshaed are equal)

The Gap (Installs minus Rights): 0

Cost to Close Gap: $0.00

Recommendation & Alternative:

Recommendation: Maintain the current compliant license posture of 30 rights for 30 installations. No additional rights need to be purchased ($0 cost).

Alternative Option: Implement continuous software metering and reclamation. If active usage monitoring shows certain users or devices haven't used ArcGIS 3D Analyst in over 90 days, reclaim those licenses to maintain a surplus buffer for future onboarding without incurring new software acquisition costs.

Part E: Track and reflection

E1: Traceability table

Requirement: Produce one table that traces the whole story with record identifiers: case, work order, tasks, faulty asset, replacement asset, both CIs, the customer contact and the software product you analysed. For each link, state which field or relationship connects the two records.

Evidence:

Module	Record Type	Record Name	Relationship	Linked To
CSM	Customer Account	CAP-AB Northwind Dental Group	Base Account Record	Sarah Jenkins (Contact)
CSM	Service Location	CAP-AB Northwind Dental Clinic	Account / Location link	CAP-AB Northwind Dental Group
CSM	Customer Contact	Sarah Jenkins	Contact / Account relationship	CAP-AB Northwind Dental Group
CSM	Case	Case record (Playbook CAP-AB Device replacement)	Account & Contact lookup	CAP-AB Northwind Dental Group & Sarah Jenkins
FSM	Work Order	Work Order generated from Case	Parent Case reference	CSM Case record
FSM	Work Order Task 1	CAP-AB Diagnose laptop	Work Order reference	Parent Work Order
FSM	Work Order Task 2	CAP-AB Replace laptop and verify	Work Order & Predecessor link	Parent Work Order & Task 1 (Predecessor)
FSM	Skills & Group	CAP-AB Hardware Diagnostics / CAP-AB Field Repair (South	Assigned Group & User Skills	Assigned Technicians

		Group / Texas Dispatchers)		
ITAM	Manufacturer	CAP-AB TechSource	Manufacturer lookup	CAP-AB Laptop 15 (Model)
ITAM	Hardware Model	CAP-AB Laptop 15	Model lookup	Hardware Assets (CAP-AB-OLD-001 & CAP-AB-NEW-001)
ITAM	Stockroom	CAP-AB Stockroom	Stockroom lookup	CAP-AB-NEW-001 (Replacement Asset)
ITAM	Faulty Asset	CAP-AB-OLD-001	Assigned to / Affected CI / Work Order Affected CI	Sarah Jenkins, Faulty CI, & Work Order
ITAM	Replacement Asset	CAP-AB-NEW-001	Assigned to / Stockroom / Work Notes	Sarah Jenkins & Work Order Work Notes
ITAM	Configuration Items	CAP-AB-OLD-001 CI & CAP-AB-NEW-001 CI	Serial Number Match Key (SN-OLD-... / SN-NEW-...)	Hardware Asset Records
ITAM	Software Product	ESRI ArcGIS 3D Analyst 10.6 (Publisher: ESRI)	Software Model & Entitlement link	Software Entitlements & Installations

E2: Reflection

Requirement: Write one page that explains how data moved from CSM to FSM to ITAM in your project, and list three improvements you would make in a real implementation. Say which of them would use capabilities you did not have in this assessment (for example HAM Pro lifecycle automation or SAM Pro reclamation).

Evidence:

In this project, data moved smoothly across ServiceNow modules to solve a hardware failure for CAP-AB Northwind Dental Group.

It all started in Customer Service Management when practice manager Sarah Jenkins reported a broken laptop at the CAP-AB Northwind Dental Clinic. When that case triggered our custom device replacement playbook, it automatically generated a Work Order in Field Service Management. This bridge passed all the critical customer and location details over to the dispatch team. The Texas Dispatchers parent group routed the work to the South assignment group, and using our custom skills, the system matched and assigned the technicians to complete the diagnostic and repair tasks in order.

Once the physical repair was done, the data moved into ITAM. The faulty asset, CAP-AB-OLD-001, was linked to the Work Order as the affected configuration item. Using the serial number match key, the asset and CMDB

records stayed completely synchronized. To finish up the hardware lifecycle, we manually retired the old laptop and reassigned the new replacement laptop from the stockroom to Sarah Jenkins as in use.

Three Improvements for a Real Implementation

1. Automated Lifecycle Workflows : we can instead build a custom workflow using core Flow Designer. This would automatically update asset states to retired or in use when a work order reaches closed complete without needing paid plugins.

2. Automated Software Monitoring using Scheduled jobs: Using a custom scheduled script execution can be configured to check installation records against last login dates and send notification emails to asset managers when software goes unused for over ninety days.

3. Automated Predictive Dispatching: While we manually assigned our custom skills and the South group, a live environment would use Advanced Work Assignment and technician geolocation to instantly dispatch the closest qualified technician the moment the case escalates.